

Practical 5 — Find and Fix

Booklet 3, pages 21-28. Uses Practicals 1-4. Success criteria 2.6, 4.3-4.5, 5.1-5.6.

Everything you write goes in your jotter. Copy each numbered table before you start that section, and answer each question by its number.

Finding a fault

An engineer's circuit does not do what it should. Find the fault the way an engineer does:

- 1 Carry out one planned test.
- 2 Compare what happens with the expected result.
- 3 Follow the chain – sensor → V_2 (V_{in}) → transistor → relay → output – to the first part that is wrong.
- 4 Change one thing.
- 5 Test again before you move on.

Write every result as a sentence: “When the ..., the”

5.1 A worked example

A motor must switch on when light falls on the LDR. Build this circuit in Yenka exactly as shown and see the initial test result for yourself.

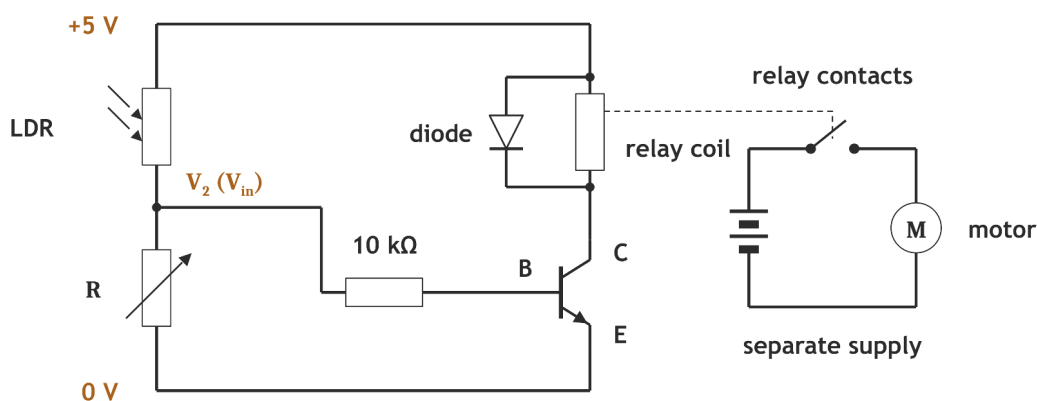


Table 5.1 — a completed test

Planned test	Expected result	Initial test result	Planned amendment	Re-test result
Shine light on the LDR.	The relay switches on and the motor turns.	When light shines on the LDR, the relay does not switch on and the motor does not turn.	Turn the protection diode round (rotate it 180°).	When light shines on the LDR, the relay switches on and the motor turns.

Q1 Explain why turning the diode round made the relay switch on.

[2]

5.2 Your turn

Build this circuit in Yenka. The motor must switch on in bright light and off in the dark, without damaging any component.

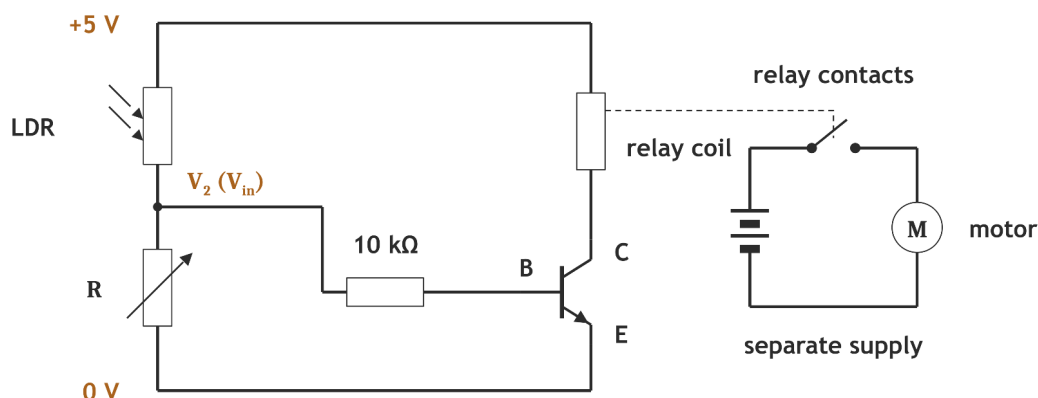


Table 5.2 – finish the test

Planned test	Expected result	Initial test result	Planned amendment	Re-test result
Shine light on the LDR, then cover it.	The motor switches on, then off. No component is damaged.	When the LDR was covered, the relay switched off and a component was destroyed.		

Q2 Copy Table 5.2 and complete the last two columns.

[2]

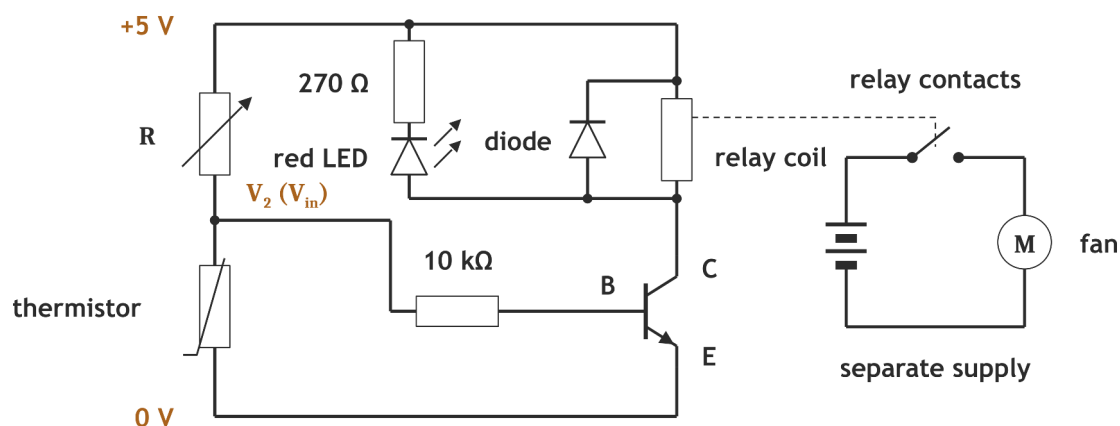
Q3 Explain why the component was destroyed when the relay switched off, and not when it switched on.

[2]

5.3 The engineer's circuit

A greenhouse fan must meet this specification:

- When the greenhouse is warm, the fan turns on and the red LED comes on.
- When the greenhouse is cool, the fan and the red LED are off.
- The fan runs from a supply separate from the 5 V control circuit.



Errors were found in this circuit during testing. Build it in Yenka exactly as shown.

5.4 Test it

Carry out the tests in order. After each amendment, test again before you move on to the next test.

Table 5.4 – testing the greenhouse circuit

Planned test	Expected result	Initial test result	Planned amendment	Re-test result
Test 1: set R so the fan is off at 10 °C, then warm the thermistor to 40 °C.	The fan turns on.			
Test 2: warm the thermistor to 40 °C.	The red LED comes on.			
Test 3: cool the thermistor to 10 °C.	The fan and the red LED switch off.			

Q4 Copy Table 5.4 and complete it for all three tests. [7]

5.5 The amended circuit

Q5 Draw your amended greenhouse circuit. [2]

5.6 Evaluate

Table 5.6 – does your amended circuit meet the specification?

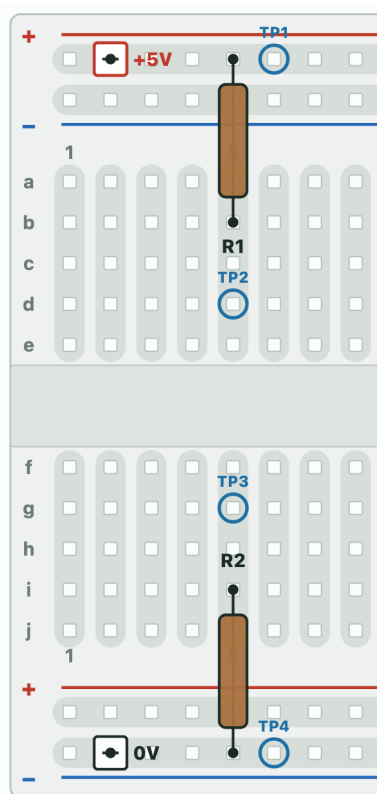
Specification point	Met? (yes / no)	Justification
i		
ii		
iii		

Q6 Copy Table 5.6 and complete it for your amended circuit. [3]

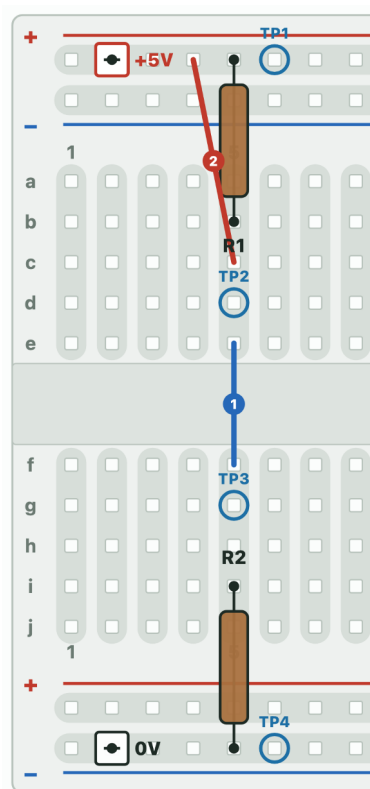
Q7 Suggest one improvement to the circuit that would benefit the gardener, and justify it. [2]

5.7 On the bench

Two pupils built Practical 1, pair A (1 k Ω and 1 k Ω) on a 5 V supply. Each board has one fault. Their meter readings are in Table 5.7.



Board D1



Board D2

Table 5.7 – meter readings

Meter between	Working board (V)	Board D1 (V)	Board D2 (V)
TP1 and TP2	2.5	0.0	0.0
TP2 and TP3	0.0	5.0	0.0
TP3 and TP4	2.5	0.0	5.0
TP1 and TP4	5.0	5.0	5.0

Q8 Board D1: identify the fault, and explain how the readings show it. [3]

Q9 Board D2: identify the fault, and explain how the readings show it. [3]

Q10 For each board, state the change that fixes the fault. [2]